Correlation Alone Does Not Identify a Judge's Conditional Rating Distribution

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Abstract

A judge can predict the correct mean rating while assigning the wrong conditional distribution to every input. Correlation between scalar judge and human scores cannot distinguish such forecasts. We argue for evaluating the conditional rating distribution with proper probability losses, alongside scalar association and agreement diagnostics. A binomial construction gives five candidate distributions with identical conditional means and distinct expected cross-entropy, Brier score and ranked probability score. With one synthetic rating per item, cross-entropy selects the true distribution in 249 of 256 replications. Pearson, Spearman, Kendall and scalar mean squared error retain all five candidates as ties throughout. We enumerate these ties, reproduce the older marginal-histogram example, and retain inconclusive historical theorem-judge evidence. The result supports distribution-aware evaluation without implying human generalization, universal finite-sample superiority, or that psychometric methods are unnecessary.

Introduction

A scalar agreement statistic does not determine whether a judge predicts how human ratings vary across inputs. G-Eval evaluates text judges using correlation with human judgments (Liu et al. 2023). MT-Bench and Chatbot Arena also study agreement with human preferences and document judge biases (Zheng et al. 2023). These comparisons answer useful questions, but do not by themselves identify a conditional rating distribution.

We take the distribution as the target when the evaluation requires probabilities for each possible rating. Proper scoring rules provide an established framework for this target (Gneiting and Raftery 2007). Our contribution is a controlled identification study that makes the information discarded by scalar criteria explicit. We do not claim a new proper scoring rule.

The study separates three questions: population identification, selection with sparse ratings, and agreement of scalar means. It compares the same candidates and splits under probability-loss and correlation selection. A separate permutation control shows why matching aggregate histograms cannot establish correct item conditioning.

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We make the following contributions:

1. We construct conditional distributions that share their means but differ under proper probability losses.
2. We evaluate selection using one, two and three synthetic ratings per item, reporting every tied minimizer and common test metric.
3. We preserve historical null findings and identify the limits of transferring this synthetic result to human-rated judges.

What Scalar Criteria Identify

Let $p(r \mid x)$ be the distribution of a specified rating population and let a judge report $q(r \mid x)$. Write $\mu_q(x) = \sum_r r q(r \mid x)$. Pearson correlation measures linear association between scalar summaries; Spearman and Kendall measure ordering with their respective tie conventions. A strictly increasing transformation preserves ranks, but compression into two bins generally introduces ties and can change rank correlation. None of these statistics identifies the full conditional distribution when different forecasts share μ_q.

Squared error also has a narrower target. For any scalar prediction $a(x)$,

$$\mathbb{E}[(a - R)^2 \mid x] = (a - \mathbb{E}[R \mid x])^2 \tag{1}$$

$$+ \operatorname{Var}(R \mid x). \tag{2}$$

It identifies the conditional mean under finite second moments, not an arbitrary conditional distribution. Scaling a bounded rating to $[0, 1]$ changes the units, not that identification property.

These limitations do not establish that every psychometric method is unnecessary. Construct validity concerns the interpretation supported by a body of evidence, rather than a single agreement coefficient (Cronbach and Meehl 1955). Choosing whose judgments to model and whether the rating task measures the intended construct remain necessary even with a proper probability loss.

Evaluate Conditional Probability Forecasts

For ratings $r \in \{0, \ldots, 5\}$ and input $x = (y, y^*)$, consider

$$\min_{\theta} \ \mathbb{E}_x\big[D_{\mathrm{KL}}(p(\cdot \mid x) \,\|\, q_\theta(\cdot \mid x))\big]. \tag{3}$$

The target p is a population distribution, not a sparse empirical histogram. Cross-entropy (CE), $\mathbb{E}_{x,R}[-\log q_\theta(R \mid x)]$,

differs from Eq. 3 by an entropy term independent of θ. One observed rating per item supplies a valid conditional log-loss observation. It does not require marginal pooling and does not identify an unconstrained distribution at that item without shared structure or further assumptions.

We also evaluate the Brier score and ranked probability score (RPS), using the loss orientation of proper scoring rules (Gneiting and Raftery 2007):

$$L_{\mathrm{Brier}}(q, r) = \sum_{j=0}^{5} (q_j - \mathbf{1}\{r = j\})^2, \tag{4}$$

$$L_{\mathrm{RPS}}(q, r) = \frac{1}{5} \sum_{j=0}^{4} (Q_j - \mathbf{1}\{r \le j\})^2, \tag{5}$$

where $Q_j = \sum_{k=0}^{j} q_k$. Appendix explains RPS in detail: intuition, propriety and why it suits ordinal human ratings. Their expected excess losses over the oracle are $\sum_j (q_j - p_j)^2$ and $\frac{1}{5} \sum_{j=0}^{4} (Q_j - P_j)^2$. These vanish only when the categorical distributions agree. CE has excess loss $D_{\mathrm{KL}}(p \| q)$, assuming forecast support covers the population support.

Empirical evaluation averages losses over observed ratings, including with one rating per input. A histogram of two or three ratings remains a noisy estimate, and pooling marginal histograms discards the input–rating association. Probability fitting also does not remove finite-sample selection error, model misspecification, or optimization failure. We retain Pearson, Spearman and Kendall as diagnostics, with normalized scalar error and signed bias when score levels matter.

Controlled Conditional Identification

Our new experiment uses $p(r \mid t) = \mathrm{Binomial}(5, t)$ for a fixed symmetric grid inside $[0.1, 0.9]$. Define

$$q_\lambda(r \mid t) = (1 - \lambda) p(r \mid t) + \lambda[(1 - t)\delta_0 + t\delta_5]. \tag{6}$$

For $\lambda \in \{0, .25, .5, .75, .95\}$, every forecast has mean $5t$. Consequently all five receive identical scalar correlations and squared errors on every realized rating dataset. Their conditional distributions differ whenever $\lambda > 0$. The excess proper losses are strictly positive; Brier and RPS excess losses scale as λ^2.

We use twelve symmetric parameter-band families, holding out six/three/three families for train/validation/test, with 96/48/48 items respectively. The nine-candidate pool adds $p(r \mid 1 - t)$, constant binomial and uniform forecasts, and a Laplace-smoothed train-rating constant. Only the final constant is fitted; every criterion selects from the same nine candidates using validation ratings. The reversed conditional preserves each split's aggregate histogram exactly by symmetry.

We generate 256 independent rating replications, each providing three ratings per item, and compare their one-, two- and three-rating prefixes. The procedure and source hashes were committed before measurement. All 768 conditions and 5,376 selections completed without missing selections. With one rating, CE selects the oracle in 249/256 replications, and Brier and RPS each select it in 245/256. Their mean exact test CE values are 1.331 [1.329, 1.334] and 1.333 [1.330, 1.336], respectively. Intervals are 95% Student-t Monte Carlo intervals across independent rating replications on fixed items; no hypothesis test was specified. All three proper losses select the oracle in 256/256 two-rating and three-rating replications, which does not prove zero selection error beyond these replications.

Candidate	CE	Brier	RPS	nMSE
q_0	1.328	.699	.104	.038
$q_{.25}$	1.446	.733	.109	.038
$q_{.50}$	1.689	.835	.125	.038
$q_{.75}$	2.164	1.004	.152	.038
$q_{.95}$	3.348	1.188	.181	.038
Reversed conditional	4.215	1.113	.384	.282

Table 1: **Identical mean predictions can have different distributional losses.** Exact expectations on the fixed synthetic test grid; CE uses nats and nMSE normalizes ratings by five. These deterministic expectations need no sampling interval.

Pearson, Spearman, Kendall and nMSE retain all five mixture candidates in every replication. We enumerate the full tie set with tolerance 10^{-12} rather than select by array order. Their exact test CE spans 1.328–3.348; uniform tie-breaking has expected CE 1.995. A tie policy that always picks the oracle attains CE 1.328, so superiority is not independent of tie policy. These are losses of individual selected forecasts, not of an averaged distribution.

The reversed conditional has identical marginal probabilities but conditional KL 2.887 nats. The five mixtures have zero normalized mean bias, mean 0.5, and normalized individual-rating MAE 0.157. Their correlations with the population conditional mean are all one, but Pearson correlation with a future individual rating is 0.786. Population-mean agreement is therefore distinct from agreement with noisy individual ratings.

Historical toy. We reproduced the original seed-0 marginal-histogram example without modifying its code or frozen output. Its noisy and two-bin judges have Pearson 0.900/0.877, Spearman 0.900/0.878, marginal KL 0.001/3.420, and nMSE 0.023/0.216. It is a single-seed marginal illustration, not conditional judge selection; the two-bin transform introduces ties.

Historical Theorem-Judge Evidence

The companion repository reports a retrospective comparison on 75 artifacts, 25 tasks, 19 families and 195 expert ratings (Trustworthy-Theorem-Judges Contributors 2026a). We import only public aggregates and preserve their limitations. Across five seeded five-fold family partitions (25 outer folds reusing those 75 artifacts), cached-histogram CE selection had held-out CE 8.868 versus 8.098 for Spearman selection. The difference was $+0.770$ $[-0.895, 1.317]$, using a 95% family-bootstrap interval that repeats selection. On the original historical split (15 validation artifacts selecting over 30 test artifacts, 10 tasks and 78 ratings), CE selection had test

CE 9.133 versus 8.710 ($+0.423$). Those historical ratings, test outcomes and candidate prompts had prior exposure, and neither denominator establishes a CE advantage.

A separate fitted-head comparison favored CE fitting by -0.370 $[-0.690, -0.080]$ in test CE, conditional on fitted heads. However, 215/225 Pearson fits failed to converge, including 24/25 final refits. This limits any objective-level interpretation of the comparison. Neither the Wasserstein nor the kernel-distance follow-up resolved an advantage over Spearman selection. All intervals are descriptive; no confirmatory hypothesis test was specified.

The separately completed weighted-probability and stable-Pearson follow-up resolved the earlier numerical comparator issue (Trustworthy-Theorem-Judges Contributors 2026b). All 50 primary fits passed checks and all 900 secondary fits converged. Its weighted-loss difference was $-0.010409\,[-0.061911, +0.005933]$, with both mixtures refitted in 998/1,000 valid family-bootstrap draws. Two numerical certificate failures were retained without replacement; the interval conditions on successful draws. It still does not establish an advantage. Our synthetic experiment neither reruns this condition nor replaces its conclusions. Fresh human returns were not independently checked from cloud for this report; we do not infer that they are absent. Natural-output generalization still requires matched probability forecasts, an explicit rater population, stable comparison procedures and appropriately held-out evidence.

Scope and Limitations

The construction deliberately includes the oracle and changes distribution shape without changing the mean. It demonstrates nonidentification by scalar summaries, not the prevalence of this failure among natural judges. The family split holds out synthetic parameter bands; it is not evidence of transfer across real tasks, solvers, raters or model families. Independent binomial ratings omit rater heterogeneity and dependence. The study trains no language model and uses no new provider calls.

One observed rating can score a conditional forecast, but learning a flexible conditional distribution requires data and shared structure. Neither a high correlation nor low finite-sample proper loss establishes that an entire human population has been learned. When only a conditional mean or ranking is required, scalar objectives may be appropriate for that narrower target. When the distribution is required, report proper probability losses and keep association, absolute error and bias as complementary diagnostics.

Conclusion

Correlation alone does not identify a judge's conditional rating distribution. Proper probability losses identify that target in the population under their assumptions; sparse data and optimization still determine empirical selection quality. Our controlled study supports this distinction while preserving inconclusive historical evidence.

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The Ranked Probability Score (RPS)

Setting. A human rating is an ordered category $r \in \{0, 1, \ldots, 5\}$. For an artifact, a judge outputs a forecast distribution $q = (q_0, \ldots, q_5)$ over ratings. Its cumulative distribution function (CDF) and survival function are

$$Q_j \stackrel{\text{def}}{=} \Pr_q[R \le j] = \sum_{k=0}^{j} q_k, \tag{7}$$

$$S_j \stackrel{\text{def}}{=} \Pr_q[R > j] = 1 - Q_j, \tag{8}$$

for thresholds $j = 0, \ldots, 4$. The last threshold is omitted because $Q_5 = 1$ for every forecast.

Definition. The ranked probability score (RPS) (Epstein 1969) compares the judge's CDF with the step function of the observed rating:

$$L_{\mathrm{RPS}}(q, r) \stackrel{\mathrm{def}}{=} \frac{1}{5} \sum_{j=0}^{4} \big(Q_j - \mathbf{1}\{r \le j\}\big)^2 \tag{9}$$

$$= \frac{1}{5} \sum_{j=0}^{4} \big(S_j - \mathbf{1}\{r > j\}\big)^2. \tag{10}$$

We use it as a loss: lower is better. Dividing by the five thresholds puts it in $[0, 1]$. It is 0 only when q puts all its mass on r, and 1 when q puts all its mass on the opposite end of the scale (for example $q_0 = 1$ when $r = 5$). Some references omit the factor $1/5$; their values are five times ours.

Empirical RPS. For N artifacts x_i with n_i human ratings $r_{i1}, \ldots, r_{in_i}$ each, we average over ratings and then artifacts:

$$\widehat{\mathrm{RPS}} = \frac{1}{N} \sum_{i=1}^{N} \frac{1}{n_i} \sum_{m=1}^{n_i} L_{\mathrm{RPS}}(q_i, r_{im}). \tag{11}$$

Writing $\hat{P}_{ij}$ for the empirical CDF of artifact i's ratings, the inner average equals $\frac{1}{5} \sum_j [(Q_{ij} - \hat{P}_{ij})^2 + \hat{P}_{ij}(1 - \hat{P}_{ij})]$. One rating per artifact already gives an unbiased estimate of the judge's expected score; no per-artifact histogram has to be estimated.

Intuition 1: five yes/no questions. At each threshold j the judge answers "is the rating at most j?" with probability Q_j, and the human rating supplies the answer. Each summand in Eq. 9 is the Brier score of one such binary forecast, so RPS is the average Brier score over the five threshold questions. Equivalently (Eq. 10), the judge forecasts the survival function: the chance that the rating exceeds each threshold.

Intuition 2: distance between staircases. The judge's CDF is a staircase that rises from 0 to 1; the observed rating is a single step at r. RPS sums the squared vertical gaps between the two at each threshold. A near miss leaves gaps at few thresholds and a far miss at many. Table 2 shows this for one human rating $r = 4$. Cross-entropy (CE) is infinite (or set by a probability floor) for every forecast with $q_4 = 0$, and the Brier score over the six categories, $\sum_k (q_k - \mathbf{1}\{r = k\})^2$, is 2 for every point mass away from 4. Only RPS gives the near miss a better (lower) score than the far miss.

Forecast for $r = 4$	RPS	Brier	CE
all mass on 4	0.000	0.0	0
all mass on 5	0.200	2.0	∞
all mass on 0	0.800	2.0	∞
half on 3, half on 5	0.100	1.5	∞

Table 2: Scores of four forecasts against one human rating $r = 4$ (lower is better).

Intuition 3: rating points. For integer ratings, RPS has an absolute-error form, the discrete case of the identity for the continuous ranked probability score (Matheson and Winkler 1976; Gneiting and Raftery 2007):

$$5\, L_{\mathrm{RPS}}(q, r) = \mathbb{E}_q |R - r| - \tfrac{1}{2} \mathbb{E}_q |R - R'|, \tag{12}$$

where R and R' are independent draws from q. The first term is the expected distance, in rating points, between a rating sampled from the judge and the human rating. The second term subtracts half the judge's own spread, so a judge is not penalized merely for spreading its probability.

Why RPS is proper. Let the human rating be drawn from p with CDF P. Then

$$\mathbb{E}_{R \sim p}\big[L_{\mathrm{RPS}}(q, R)\big] = \underbrace{\frac{1}{5} \sum_{j=0}^{4} (Q_j - P_j)^2}_{\text{excess: judge's error}} + \underbrace{\frac{1}{5} \sum_{j=0}^{4} P_j (1 - P_j)}_{\text{rater disagreement}}. \tag{13}$$

The second term does not depend on q: it is the irreducible disagreement among raters. The first term is nonnegative and is zero only when $Q = P$, that is, $q = p$. So RPS is strictly proper (Gneiting and Raftery 2007): a judge minimizes its expected RPS only by reporting the full human rating distribution, including how much raters disagree.

Example: disagreement that correlation cannot see. Suppose raters split evenly between ratings 3 and 5 on an artifact. Judge A predicts the human distribution; Judge B puts all its mass on 4. Both have expected rating 4, so on every such artifact they produce the same expected rating and any correlation with human mean ratings scores them identically. Their expected RPS differs: 0.10 for Judge A (all irreducible disagreement) and 0.20 for Judge B. A uniform forecast over $0, \ldots, 5$ scores 0.206.

Why the cumulative form: robustness with few ratings. Each artifact has only two or three human ratings, so its rating histogram leaves at least three of the six bins empty. Distances computed bin by bin between the judge's probability mass function and such a histogram are dominated by the empty bins. The Kullback–Leibler divergence is infinite or set by a probability floor. Total variation distance (TVD) is biased upward even for the true distribution. Suppose raters choose 3 with probability 0.6 and 5 with probability 0.4. The true distribution's expected TVD to its own histogram of 1, 2 or 3 ratings is 0.48, 0.288 and 0.23. With one rating per artifact, a forecast that puts all its mass on 3 scores better (0.40) than the truth (0.48), so plug-in TVD rewards overconfidence. Scoring each rating by the absolute gap between CDFs, the one-rating Wasserstein-1 distance, has the same flaw (0.80 for the point mass vs 0.96 for the truth) because it is minimized by a point mass at the median. RPS ranks the truth first (0.096 vs 0.160).

The cumulative form avoids the problem. Each $Q_j = \Pr[R \le j]$ pools every rating category at or below j, so every

rating informs every threshold and no threshold is ever empty. The empirical CDF at a threshold is an average of indicators and therefore an unbiased estimate of P_j. Its sampling noise contributes exactly the judge-independent term $P_j(1 - P_j)$ in Eq. 13, which cancels when two judges are compared on the same artifacts. Squaring the gaps, rather than taking their absolute values, is what makes the score proper. A related cumulative view appears in Schaeffer et al. (2024), who compared distributions of per-sample score–compute correlations through survival functions (complementary CDFs). Those read off the fraction of samples above any threshold, and the authors summarized them with the area under the survival function and with Wasserstein distances, which are also defined through CDFs.

RPS as a distance between distributions, and "root RPS". If the human distribution p were known, the excess term in Eq. 13, $\frac{1}{5}\sum_j (Q_j - P_j)^2$, would be a squared ℓ_2 distance between the two CDFs. This is the squared Cramér distance (Bellemare et al. 2017), and on the integer rating scale it equals one tenth of the energy distance between q and p (Székely and Rizzo 2013). Its square root is a true metric between rating distributions, a principled "root RPS". In practice p is unknown, so we estimate the expected RPS from observed ratings. For one artifact the excess and the disagreement floor cannot be separated, but the floor is the same for every judge. Differences in mean RPS between two judges are therefore unbiased estimates of differences in their squared Cramér distances to the human distributions (up to the factor $\frac{1}{5}$). We know of no standard metric called root RPS. Taking the square root of a mean RPS does not change how judges are ranked, but taking it per rating before averaging would lose this unbiasedness. Bellemare et al. (2017) also show that, unlike the Wasserstein distance, the Cramér distance has unbiased sample gradients, which makes RPS usable as a training loss as well as an evaluation metric.

Why RPS suits judges of ordinal human ratings.

- **Order matters.** Ratings are ordered; RPS charges more for larger misses. CE and the categorical Brier score treat ratings as unordered labels.
- **It targets the whole distribution.** Being strictly proper, RPS is minimized only by the true conditional rating distribution. Correlation ignores calibration and spread, and mean squared error identifies only the conditional mean.
- **It works with sparse ratings.** Each observed rating is scored directly through the CDF, so artifacts with one to three ratings contribute without estimating a histogram, and empty rating categories do not distort the score.
- **It is bounded.** RPS lies in $[0, 1]$ and cannot be dominated by one rating that received near-zero probability. CE can: in our retrospective human-rating analysis (archived experiment 11), judge mixtures had higher held-out CE than a constant forecast of the training ratings (5.43 vs 1.85 nats), driven by the 10^{-6} probability floor, while their held-out RPS was lower (0.137 vs 0.183).
- **It is interpretable.** The absolute-error form is in rating points, and Eq. 13 separates the judge's error from the noise floor set by rater disagreement.

Limitations. RPS assumes equally spaced ratings: a miss from 0 to 1 costs the same as a miss from 4 to 5. It weights all thresholds equally. It is less sensitive than CE to the probability given to rare ratings, which is why we keep CE as a secondary metric. With two or three ratings per artifact, $\hat{P}_{ij}(1 - \hat{P}_{ij})$ is a noisy estimate of rater disagreement, so comparisons between judges should use paired differences on the same artifacts. Galdran (2023) notes two further behaviors. First, the penalty for a confident miss grows only linearly with distance: a point mass d ratings away from r scores $d/5$. Second, RPS can prefer a symmetric forecast to one with more probability on the observed rating. That work proposes a squared-absolute variant, $\frac{1}{5}\left(\sum_j |Q_j - \mathbf{1}\{r \le j\}|\right)^2$. In our numeric check this variant is not proper. For raters distributed $(0.2, 0.2, 0.2, 0.2, 0.1, 0.1)$ over ratings $0, \ldots, 5$, the forecast $(0, 0, 0.9, 0.1, 0, 0)$ has a lower expected variant score (0.498) than the true distribution (0.680), whereas RPS prefers the truth (0.178 vs 0.246). We therefore keep RPS.